

Structured Data Collection Plan for an Object Counting Survey Paper

Focus: object counting across modalities - images, videos, and emerging settings such as 3D/point clouds, remote sensing, microscopy, and language-guided/open-vocabulary counting.

1. Research Goal and Scope

The goal is not to manually review every paper returned by Google Scholar. The goal is to build a controlled, reproducible seed set of important papers, then expand and organize the literature using scholarly APIs and manual screening. The survey should answer: **How has object counting evolved across modalities, and what methods, datasets, prompts, and evaluation metrics define each modality?**

Proposed survey title: Object Counting Across Modalities: A Survey of Image, Video, Open-Vocabulary, and Emerging 3D Counting Methods.

2. Overall Workflow

Phase	Purpose	Main output
Phase 1: Manual seed discovery	Use Google Scholar only to find high-quality seed papers, not to scrape thousands of results. Search by modality and task type.	55-85 seed papers with titles, BibTeX, PDFs/links, and short notes.
Phase 2: Automated expansion	Use Claude Code/Codex with OpenAlex, Semantic Scholar, Crossref, arXiv, and exported BibTeX to collect metadata, references, citations, and related papers.	A larger candidate pool, deduplicated by DOI and fuzzy title matching.
Phase 3: Screening	Filter titles and abstracts using inclusion/exclusion rules. Assign each paper to modality and method categories.	A screened literature matrix of about 100-150 relevant papers.
Phase 4: Deep reading	Read the most important papers in detail. Extract method, input modality, datasets, metrics, limitations, and relevance.	40-80 deeply read papers for the final survey narrative.
Phase 5: Writing assets	Generate taxonomy, tables, timeline, dataset summary, and BibTeX file.	Survey outline, references.bib, literature_matrix.xlsx, and taxonomy tables.

3. Target Seed Paper Counts

Bucket	Target	What to include
Survey/taxonomy papers	5-10	Existing surveys, taxonomy papers, benchmark overviews, and review-style papers.
Image object counting	20-30	Few-shot, exemplar-based, class-agnostic, zero-shot, open-vocabulary, SAM/foundation-model counting.
Video object counting	10-15	Video counting, open-world video counting, tracking-based counting, re-identification, TAO-Count, VideoCount, CountVid.
Other modalities/domains	10-15	3D/point-cloud, RGB-D, remote sensing, UAV, microscopy/cell counting, X-ray, thermal, event camera, science videos.
Datasets and benchmarks	10-15	FSC-147, FSCD-147, CountBench, OmniCount, CARPK, ShanghaiTech, TAO-Count, VideoCount, MOT20-Count, Science-Count.
Total	55-85	Balanced seed library before automated expansion.

4. Google Scholar Search Strategy

Use Google Scholar as a seed discovery and sanity-check tool. For each query, inspect only the top 20 results by relevance, the top 10 results by date, and the "Cited by" / "Related articles" links for the strongest papers. Stop after the target count is reached.

Modality/task bucket	Google Scholar queries to run	Seed target
Surveys and taxonomy	"class-agnostic counting" survey "object counting" "computer vision" survey "visual counting" survey "few-shot object counting" survey "open-vocabulary object counting" survey	5-10
Image counting	"few-shot object counting" "exemplar-based object counting" "class-agnostic object counting" "zero-shot object counting" "open-vocabulary object counting" "object counting" "FSC-147" "object counting" "SAM" "visual counting" "vision language models"	20-30
Video counting	"video object counting" "open-world object counting in videos" "object counting in videos" "video counting" tracking "video counting" re-identification "unique instance counting" video "TAO-Count" object counting "VideoCount" object counting "CountVid" object counting	10-15
Other modalities	"3D object counting" "object counting" "point cloud" "RGB-D object counting" "remote sensing" "object counting" "aerial image" "object counting" "UAV" "object counting" "microscopy" "object counting" "cell counting" "computer vision" "thermal image" "object counting" "event camera" "object counting"	10-15
Datasets/benchmarks	"FSC-147" object counting "FSCD-147" object counting "CountBench" object counting "OmniCount" object counting "CARPK" counting "ShanghaiTech" counting "TAO-Count" "VideoCount" object counting "MOT20-Count" "Science-Count"	10-15

5. Seed Paper Collection Rules

Inclusion criteria:

Keep a paper if it is directly about object counting, visual counting, counting benchmarks, or a modality/domain where counting is the central output. Also keep papers that introduce major datasets, metrics, or evaluation protocols for counting.

Exclusion criteria:

Remove papers that only mention counting as a minor application, are mostly generic object detection without count evaluation, are unrelated to images/videos/visual modalities, or are too domain-specific without providing transferable insights.

Ranking rule:

Prioritize papers that are recent, highly cited, from strong venues, dataset-defining, method-defining, survey/taxonomy papers, or directly related to your target focus: image counting, video counting, zero-shot/open-vocabulary counting, and foundation-model-based counting.

6. Literature Matrix Columns

Column	Purpose
Paper title	Exact title for deduplication and citation lookup.
Authors / year / venue	Basic bibliographic information.
DOI / arXiv / URL	Stable identifiers and access links.
Modality	Image, video, 3D, point cloud, RGB-D, remote sensing, microscopy, X-ray, thermal, event camera, or other.
Task type	Density-based, detection-based, exemplar-based, few-shot, zero-shot, class-agnostic, open-vocabulary, tracking-based, or foundation-model-based.
Input	Image, video, text prompt, exemplars, boxes, masks, depth, point cloud, or multimodal input.
Output	Count only, boxes + count, masks + count, tracks + count, density map, or unique instance count.
Datasets	Benchmarks used in the paper.
Metrics	MAE, RMSE, NAE, SRE, AP/AR, count accuracy, tracking identity metrics, or domain-specific metrics.
Key contribution	One or two sentences describing the main novelty.
Limitations	Known weaknesses, modality limitations, scalability issues, or evaluation gaps.
Relevance score	High, medium, low. Helps decide what to read deeply.
BibTeX copied? PDF saved?	Yes/no tracking fields for reference management.

7. Automated Expansion with Claude Code or Codex

After the seed set is collected, use Claude Code or Codex to build a reproducible pipeline. The automation should use legitimate scholarly data sources rather than bulk scraping Google Scholar. Recommended data sources include OpenAlex, Semantic Scholar, Crossref, arXiv, and manually exported BibTeX from Google Scholar/Zotero.

Prompt to give Claude Code/Codex:

Create a Python literature-review pipeline for a survey paper on object counting across modalities.

Input files:

- seed_papers.csv with title, authors, year, DOI/arXiv if available, modality, and notes.
- queries.txt with search queries grouped by modality.

Use OpenAlex, Semantic Scholar, Crossref, and arXiv APIs to retrieve:

- title, authors, year, venue, DOI/arXiv, URL
- abstract
- citation count if available
- references and related papers where available
- BibTeX metadata where available

Then:

1. Deduplicate by DOI, arXiv ID, and fuzzy title matching.
2. Classify each paper into modality and task categories.
3. Score relevance as high, medium, or low based on title/abstract keywords.
4. Save outputs as:
 - literature_matrix.xlsx
 - references.bib
 - grouped_papers.md
 - survey_outline.md
 - screening_log.csv
5. Add clear comments so the pipeline can be rerun and updated later.

8. Screening Stages

Stage	Action	Decision
Title screening	Quickly remove clearly unrelated papers.	Keep / reject / unsure.
Abstract screening	Read abstracts for papers marked keep or unsure.	Assign modality, task type, and relevance score.
Full-paper screening	Read methods, datasets, experiments, and limitations.	Decide whether paper belongs in main narrative, related work table, or background only.
Final synthesis	Group papers into taxonomy and identify trends/gaps.	Use 40-80 papers deeply in final survey.

9. Proposed Taxonomy for the Survey

Axis	Categories
Modality	Image, video, 3D/point cloud, RGB-D, remote sensing/aerial, microscopy/medical, thermal/event camera, scientific imagery.
Input supervision	Class labels, text prompts, exemplars, bounding boxes, masks, no exemplars, multimodal prompts.
Counting paradigm	Density-map regression, detection-based counting, segmentation-based counting, exemplar matching, zero-shot/open-vocabulary counting, tracking-based unique counting.
Output type	Total count, localized boxes, masks, density maps, object tracks, unique object identities.
Evaluation	MAE, RMSE, NAE, SRE, count accuracy, AP/AR, identity consistency, FPS/runtime, memory.

10. Five-Day Practical Plan

Day	Task	Deliverable
Day 1	Collect survey/taxonomy papers and define final buckets.	5-10 survey/taxonomy seeds; first version of taxonomy.
Day 2	Collect image-counting papers: few-shot, exemplar-based, class-agnostic, zero-shot, open-vocabulary, foundation-model/SAM-related.	20-30 image-counting seeds.
Day 3	Collect video-counting papers: video object counting, open-world video counting, tracking/re-ID, CountVid, VideoCount, TAO-Count.	10-15 video-counting seeds.

Day	Task	Deliverable
Day 4	Collect emerging modalities and dataset/benchmark papers.	20-30 papers across 3D, remote sensing, microscopy, datasets, and benchmarks.
Day 5	Run Claude Code/Codex pipeline for expansion, deduplication, classification, and spreadsheet creation.	literature_matrix.xlsx, references.bib, grouped_papers.md, survey_outline.md.

11. Final Deliverables from the Data Collection Phase

By the end of data collection, you should have: **(1)** a seed paper spreadsheet, **(2)** an expanded and deduplicated literature matrix, **(3)** a clean BibTeX file, **(4)** grouped papers by modality/task type, **(5)** a survey outline, **(6)** a dataset/benchmark table, and **(7)** a reading priority list of high-relevance papers.

Important rule: Google Scholar is for seed discovery and verification. Claude Code/Codex plus scholarly APIs are for bulk organization, metadata retrieval, deduplication, and literature-matrix generation.

12. Recommended First Step

Create a spreadsheet named **object_counting_seed_papers.xlsx** with the columns listed above. Then run only the survey/taxonomy queries first. Once the first 5-10 papers are collected, use their references and related articles to refine the rest of the search. This prevents the survey from becoming too broad and helps keep the paper centered on modalities, methods, datasets, and evaluation metrics.